

Khalil Lagha

Seeking an apprenticeship - power electronics and electrical energy systems

+33698347550 - Lkhalilellah@gmail.com - Authorized to work in France

linkedin.com/in/khalil-ellah-lagha - github.com/KhalilEllahLAGHA - khalilellahlagha.github.io

SUMMARY

M2 student in Electrical Energy Systems Design (CSEE, UGA), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **power electronics: DC-DC converters (buck, boost)**, AC-DC rectification, **regulation-loop tuning (PI/PID)**, **efficiency optimization**, EMI compliance - validated in simulation (Matlab/Simulink) then on real hardware, across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain, electrical grids, AI applied to power systems**. **Seeking: work-study contract (alternance) from September 2026 - 2-3-week company/university rotation.**

EXPERIENCE

Research Intern - GIPSA-lab

(04/2026 - 07/2026, 3.5 months)

- Modeled a **10-cell PEMFC fuel-cell stack** in Matlab/Simulink and achieved **6%** dynamic error and **1%** static error against real measurements
- Tuned a **DC-DC boost converter PI regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench

Intern - SLB (Schlumberger)

(07/2025 - 07/2025, 1 month)

- Authored technical syntheses **in English** (directional drilling) in an international industrial environment

Power Electronics & Automation Intern - EURL Lagha

(01/2024 - 01/2024, 15 days - 09/2024 - 09/2024, 1 month)

- Contributed to an **industrial DC-DC buck converter** - 50 V vers adjustable 0-48 V, **25 A**
- Contributed to **4 transformer-rectifiers** feeding pipelines (electrolysis) - three-phase 400 V vers 100 V transformer, rectification + capacitive filtering: **100 V DC / 80 A (environ 8 kW)** output, ripple moins de 5%, **EMI compliance**
- Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC)
- Analyzed a customer specification and sized the corresponding transformer; produced the electrical schematics in **EPLAN**

Grid Intern - Sonelgaz

(03/2024 - 03/2024, 15 days, observation)

- Analyzed the full generation vers distribution chain (**EHV 400/220 kV vers HV 60 kV vers MV 30/10 kV vers LV 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit

PROJECTS

- **Power grid analysis** (Matlab, solo) - developed **load flows from 3 to 118 buses** (PQ/PV/slack; IEEE 14/30/57/118 networks); benchmarked **3 numerical methods** - Gauss-Seidel, Newton-Raphson, fast-decoupled (accuracy vs computation time); studied **transient stability**
- **Field-oriented control (FOC) of a 5 kW / 380 V induction motor** (Python + Matlab/Simulink) - rotor-flux orientation, cascaded PI, space-vector PWM, flux estimator; **three matching implementations** incl. Simscape Electrical at 10 kHz (open source)
- **Electrical machine sizing** (Python, team) - built a **Python (OOP, PyQt)** application: stator winding design, flux density via reluctance networks, **FEMM** finite-element validation, automatic parameter generation
- **Neural network from scratch (MLP)** (Matlab, open source) - hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; published under the MIT license - github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab
- Also: cascade control of a DC motor (three-phase thyristor bridge, Simulink); Siemens Step7 automation project (HMI + Ladder); Arduino line-follower robot (Poly Maze competition)

EDUCATION

- **M2 CSEE - Electrical Energy Systems Design**, Universite Grenoble Alpes - from September 2026, apprenticeship
- **M1 EEA (EECS)**, Universite Grenoble Alpes (2025-2026) - **ranked 2nd of the cohort, annual average 15.43/20**
- **Engineering cycle in Electrical Engineering**, Ecole Nationale Polytechnique, Algiers (2021-2025) - 4 of 5 years completed; prep classes: ranked 16th/358 (year 1), 49th/264 (year 2)
- **Baccalaureat in Mathematics** (2021) - highest honors (16.93/20)

SKILLS

- **Power electronics:** DC-DC conversion (buck, boost), AC-DC (diode and thyristor rectification), DC-AC, AC-AC; converter sizing; PWM; filtering, ripple; EMI compliance; efficiency optimization
- **Control engineering:** regulation-loop tuning (PI/PID), LQR, observers, Kalman filter; field-oriented control (FOC); cascade control; system identification
- **Machines & grids:** electrical machines and transformers (analysis, design); power system analysis - load flow, Y/Z matrices, transient stability; MV/LV distribution
- **Industrial automation:** Siemens PLC (TIA Portal, Step7 - Ladder, SCL, FBD, SFC); HMI/SCADA; Grafcet
- **AI:** neural networks (backpropagation), genetic algorithms, fuzzy logic
- **Software:** Matlab/Simulink (Simscape Power Systems) - Python (OOP, PyQt) - C/C++ - LTspice - Proteus - EPLAN - AutoCAD Electrical - Siemens TIA Portal - Step7 (Ladder/SCL) - FEMM - SolidWorks - Arduino - Git/GitHub

SOFT SKILLS

Fast learner - Comfortable with tools/software - Problem solving - Rigorous - Organized - IEEE Student Branch ENP - VIC (ENP)

LANGUAGES

French - C1 - **English - C2**

CERTIFICATIONS

- SLB - NEST & SIPP Level 2 (07/2025) - HSE, industrial-operations safety